

- Module ISO8823-PRESENTATION (X.226:07/1994)
- See also [ITU-T X.226 \(1994\) Amend. 2 \(12/1997\)](#)
- See also the [index](#) of all ASN.1 assignments needed in this document

ISO8823-PRESENTATION **DEFINITIONS** ::= **BEGIN**

```
--
-- In X.410-1984 mode, the value of the SS-user data parameter of the S-CONNECT request and
-- indication session-service primitives shall be a CP-type value.
--
-- In normal mode, the value of the SS-user data parameter of the S-CONNECT request and
-- indication session-service primitives shall be a CP-type value, followed as a requestor's
-- option by zero or more CPC-type values.
IMPORTS
  RTORQapdu, RTOACapdu, RTORJapdu, RTABapdu
  FROM Reliable-Transfer-APDU {joint-iso-itu-t(2) reliable-transfer(3)
    apdus(0)};

--
CP-type ::= SET {
  mode-selector          [0] IMPLICIT Mode-selector,
  x410-mode-parameters
    [1] IMPLICIT SET {COMPONENTS OF Reliable-Transfer-APDU.RTORQapdu} OPTIONAL-- This OPTIONAL element shall be absent for a
-- nested presentation connection.
-- Shall be used for X.410 mode only. Shall be bitwise
-- compatible with CCITT Recommendation X.410-1984.
-- This shall be the User data parameter of the CP PDU1) --,
  normal-mode-parameters
    [2] IMPLICIT SEQUENCE {protocol-version
      [0] IMPLICIT Protocol-version DEFAULT {version-1},
      calling-presentation-selector
      [1] IMPLICIT Calling-presentation-selector
        OPTIONAL,
      called-presentation-selector
      [2] IMPLICIT Called-presentation-selector OPTIONAL,
      presentation-context-definition-list
      [4] IMPLICIT Presentation-context-definition-list
        OPTIONAL,
      default-context-name
      [6] IMPLICIT Default-context-name OPTIONAL,
      presentation-requirements
      [8] IMPLICIT Presentation-requirements OPTIONAL,
      user-session-requirements
      [9] IMPLICIT User-session-requirements OPTIONAL-- shall not be present if equal to the Revised session
```

```

-- requirements parameter --,
protocol-options
  [11] Protocol-options DEFAULT {},
-- shall be absent if no options are offered
initiators-nominated-context
  [12] Presentation-context-identifier OPTIONAL,
-- shall only be present if nominated-context is
-- offered in protocol-options
extensions
  [14] SEQUENCE {...} OPTIONAL,
-- to allow for future extensions
user-data
  User-data OPTIONAL} OPTIONAL
-- Shall be used for normal mode only.
-- Shall be the parameters of the CP PPDU.
}

--
-- As an initiator's option, the presentation data values contained in a CP PPDU may be
-- encoded more than once, using CPC-type values, to allow the transfer of the same
-- presentation data values using a number of different transfer syntaxes.
--
CPC-type ::=
  User-data

-- Shall be used for normal mode only.
-- Shall not be present if the Presentation context definition list parameter is not present
-- in the CP PPDU. Each instance of this data type shall contain all of the presentation data
-- values which were contained in the User data parameter of the CP PPDU.
-- This shall be the same set of presentation data values which were contained in the CP-type.
--
-- The SS-user data parameter value of the S-CONNECT response
-- and confirm session-service primitives shall be a CPA-PPDU
-- value when the Result parameter value is "accept".
--
CPA-PPDU ::= SET {
  mode-selector          [0] IMPLICIT Mode-selector,
  x410-mode-parameters
    [1] IMPLICIT SET {COMPONENTS OF Reliable-Transfer-APDU.RT0ACapdu} OPTIONAL-- This OPTIONAL element shall be absent for a
    -- nested presentation connection.
    -- Shall be used for X.410 mode only. Shall be bitwise
    -- compatible with CCITT Recommendation X.410-1984.
    -- This shall be the User data parameter of the CPA PPDU1) --,
  normal-mode-parameters
    [2] IMPLICIT SEQUENCE {protocol-version
      [0] IMPLICIT Protocol-version DEFAULT {version-1},

```

```

responding-presentation-selector
  [3] IMPLICIT Responding-presentation-selector
    OPTIONAL,
presentation-context-definition-result-list
  [5] IMPLICIT Presentation-context-definition-result-list
    OPTIONAL,
presentation-requirements
  [8] IMPLICIT Presentation-requirements OPTIONAL,
user-session-requirements
  [9] IMPLICIT User-session-requirements OPTIONAL,
  -- shall not be present if equal to the Revised session
  -- requirements parameter
protocol-options
  [11] Protocol-options DEFAULT {},
  -- shall be absent if no options are selected
responders-nominated-context
  [13] Presentation-context-identifier OPTIONAL,
  -- shall only be present if nominated-context is
  -- selected in protocol-options
user-data
  User-data OPTIONAL} OPTIONAL
-- Shall be used for normal mode only.
}

--
--
-- The SS-user data parameter value of the S-CONNECT response
-- and confirm session-service primitives shall be a CPR-PPDU
-- value when the Result parameter value is "reject by SS-provider"
-- or "reject by called SS-user".
--
CPR-PPDU ::= CHOICE {
  x400-mode-parameters SET {COMPONENTS OF Reliable-Transfer-APDU.RTORJapdu}-- This OPTIONAL element shall be absent for a
  -- nested presentation connection.
  -- Shall be used for X.410 mode only. Shall be bitwise
  -- compatible with CCITT Recommendation X.410-1984.
  -- This shall be the User data parameter of the CPR PPDU2) --,
  normal-mode-parameters
    SEQUENCE {protocol-version
      [0] IMPLICIT Protocol-version DEFAULT {version-1},
      responding-presentation-selector
      [3] IMPLICIT Responding-presentation-selector OPTIONAL,
      presentation-context-definition-result-list
      [5] IMPLICIT Presentation-context-definition-result-list
        OPTIONAL,
      default-context-result
      [7] IMPLICIT Default-context-result OPTIONAL,

```

```

    provider-reason
    [10] IMPLICIT Provider-reason OPTIONAL,
    user-data User-data OPTIONAL
}
-- Shall be used for normal mode only.
}

--
--
-- The SS-user data parameter of the S-U-ABORT request and indication
-- service primitives shall be an Abort-type value.
--
Abort-type ::= CHOICE {
    aru-ppdu ARU-PPDU -- for a P-U-ABORT -- ,
    arp-ppdu ARP-PPDU -- for a P-P-ABORT
}

--
ARU-PPDU ::= CHOICE {
    x400-mode-parameters SET {COMPONENTS OF Reliable-Transfer-APDU.RTABapdu} -- Shall be used for X.410 mode only. Shall be bitwise
    -- compatible with CCITT Recommendation X.410-1984.
    -- This shall be the User data parameter of the ARU PPDU3. --,
    normal-mode-parameters
        [0] IMPLICIT SEQUENCE {presentation-context-identifier-list
            [0] IMPLICIT Presentation-context-identifier-list
            OPTIONAL,
            user-data
            User-data OPTIONAL}
    -- Shall be used for normal mode only.
}

--
ARP-PPDU ::= SEQUENCE {
    provider-reason [0] IMPLICIT Abort-reason OPTIONAL,
    event-identifier [1] IMPLICIT Event-identifier OPTIONAL
}

--
--
-- The SS-user data parameter value of the S-TYPED-DATA request
-- and indication service primitives shall be a Typed-data-type value.
--
Typed-data-type ::= CHOICE {
    acPPDU [0] IMPLICIT AC-PPDU -- P-ALTER-CONTEXT request and indication -- ,
    acaPPDU [1] IMPLICIT ACA-PPDU -- P-ALTER-CONTEXT response and confirm -- ,
    ttdPPDU User-data -- P-TYPED-DATA request and indication
}

```

```

--
AC-PPDU ::= SEQUENCE {
    presentation-context-addition-list
        [0] IMPLICIT Presentation-context-addition-list OPTIONAL,
    presentation-context-deletion-list
        [1] IMPLICIT Presentation-context-deletion-list OPTIONAL,
    user-data User-data OPTIONAL
}

--
ACA-PPDU ::= SEQUENCE {
    presentation-context-addition-result-list
        [0] IMPLICIT Presentation-context-addition-result-list OPTIONAL,
    presentation-context-deletion-result-list
        [1] IMPLICIT Presentation-context-deletion-result-list OPTIONAL,
    user-data User-data OPTIONAL
}

--
-- The SS-user data parameter value of the S-RESYNCHRONIZE request and
-- indication service primitives shall be an RS-PPDU value.
--
RS-PPDU ::= SEQUENCE {
    presentation-context-identifier-list
        [0] IMPLICIT Presentation-context-identifier-list OPTIONAL,
    user-data User-data OPTIONAL
}

--
-- The SS-user data parameter value of the S-RESYNCHRONIZE response and
-- confirm service primitives shall be an RSA-PPDU value.
--
RSA-PPDU ::= SEQUENCE {
    presentation-context-identifier-list
        [0] IMPLICIT Presentation-context-identifier-list OPTIONAL,
    user-data User-data OPTIONAL
}

--
-- The SS-user data parameter values of the S-DATA, S-CAPABILITY-DATA,
-- S-EXPEDITED-DATA request and indication session-service primitives
-- and S-CAPABILITY-DATA response and confirm session-service primitives
-- shall be of type User-data.
--

```

*-- The SS-user data parameter values of all other session-service
primitives not described above shall be of type User-data.*

```

---
---
Abort-reason ::= INTEGER {
    reason-not-specified(0), unrecognized-ppdu(1), unexpected-ppdu(2),
    unexpected-session-service-primitive(3), unrecognized-ppdu-parameter(4),
    unexpected-ppdu-parameter(5), invalid-ppdu-parameter-value(6)}

```

Abstract-syntax-name ::= **OBJECT IDENTIFIER**

Called-presentation-selector ::= [Presentation-selector](#)

Calling-presentation-selector ::= [Presentation-selector](#)

```

Context-list ::=
    SEQUENCE SIZE (0..7, ..., 8..MAX) OF
        SEQUENCE {presentation-context-identifier Presentation-context-identifier,
                    abstract-syntax-name Abstract-syntax-name,
                    transfer-syntax-name-list SEQUENCE OF Transfer-syntax-name
        }

```

```

Default-context-name ::= SEQUENCE {
    abstract-syntax-name [0] IMPLICIT Abstract-syntax-name,
    transfer-syntax-name [1] IMPLICIT Transfer-syntax-name
}

```

Default-context-result ::= [Result](#)

```

Event-identifier ::= INTEGER {
    cp-PPDU(0), cpa-PPDU(1), cpr-PPDU(2), aru-PPDU(3), arp-PPDU(4), ac-PPDU(5),
    aca-PPDU(6), td-PPDU(7), ttd-PPDU(8), te-PPDU(9), tc-PPDU(10), tcc-PPDU(11),
    rs-PPDU(12), rsa-PPDU(13), s-release-indication(14), s-release-confirm(15),
    s-token-give-indication(16), s-token-please-indication(17),
    s-control-give-indication(18), s-sync-minor-indication(19),
    s-sync-minor-confirm(20), s-sync-major-indication(21),
    s-sync-major-confirm(22), s-p-exception-report-indication(23),
    s-u-exception-report-indication(24), s-activity-start-indication(25),
    s-activity-resume-indication(26), s-activity-interrupt-indication(27),
    s-activity-interrupt-confirm(28), s-activity-discard-indication(29),
    s-activity-discard-confirm(30), s-activity-end-indication(31),
    s-activity-end-confirm(32)}

```

```

Mode-selector ::= SET {
    mode-value [0] IMPLICIT INTEGER {x410-1984-mode(0), normal-mode(1)}
}

```

```

Presentation-context-addition-list ::= Context-list

Presentation-context-addition-result-list ::= Result-list

Presentation-context-definition-list ::= Context-list

Presentation-context-definition-result-list ::= Result-list

Presentation-context-deletion-list ::=
    SEQUENCE SIZE (0..7, ..., 8..MAX) OF Presentation-context-identifier

Presentation-context-deletion-result-list ::=
    SEQUENCE SIZE (0..7, ..., 8..MAX) OF
        INTEGER {acceptance(0), user-rejection(1)}

Presentation-context-identifier ::= INTEGER(1..127, ..., 128..MAX)

Presentation-context-identifier-list ::=
    SEQUENCE SIZE (0..7, ..., 8..MAX) OF
        SEQUENCE {presentation-context-identifier Presentation-context-identifier,
            transfer-syntax-name Transfer-syntax-name}

Presentation-requirements ::= BIT STRING {context-management(0), restoration(1)}

Presentation-selector ::= OCTET STRING(SIZE (1..4, ..., 5..MAX))

Protocol-options ::= BIT STRING {
    nominated-context(0), short-encoding(1), packed-encoding-rules(2)}

Protocol-version ::= BIT STRING {version-1(0)}

Provider-reason ::= INTEGER {
    reason-not-specified(0), temporary-congestion(1), local-limit-exceeded(2),
    called-presentation-address-unknown(3), protocol-version-not-supported(4),
    default-context-not-supported(5), user-data-not-readable(6),
    no-PSAP-available(7)}

Responding-presentation-selector ::= Presentation-selector

Result ::= INTEGER {acceptance(0), user-rejection(1), provider-rejection(2)}

Result-list ::=
    SEQUENCE SIZE (0..7, ..., 8..MAX) OF
        SEQUENCE {result [0] IMPLICIT Result,
            transfer-syntax-name [1] IMPLICIT Transfer-syntax-name OPTIONAL,

```

```

    provider-reason
      [2] IMPLICIT INTEGER {reason-not-specified(0),
                           abstract-syntax-not-supported(1),
                           proposed-transfer-syntaxes-not-supported(2),
                           local-limit-on-DCS-exceeded(3)} OPTIONAL
  }

Transfer-syntax-name ::= OBJECT IDENTIFIER

User-data ::= CHOICE {
  simply-encoded-data  [APPLICATION 0] IMPLICIT Simply-encoded-data,
  fully-encoded-data   [APPLICATION 1] IMPLICIT Fully-encoded-data,
  ...
}

-- Subclause 8.4 defines when each of the two alternatives shall be used.
Simply-encoded-data ::= OCTET STRING

-- See 8.4.1.
Fully-encoded-data ::= SEQUENCE SIZE (1, ..., 2..MAX) OF PDV-list

-- contains one or more PDV-list values.
-- See 8.4.2.
PDV-list ::= SEQUENCE {
  transfer-syntax-name      Transfer-syntax-name OPTIONAL,
  presentation-context-identifier Presentation-context-identifier,
  presentation-data-values
    CHOICE {single-ASN1-type
      [0] ABSTRACT-SYNTAX.&Type
        (CONSTRAINED BY {
          -- Type corresponding to presentation context identifier -- }),
      octet-aligned [1] IMPLICIT OCTET STRING,
      arbitrary [2] IMPLICIT BIT STRING}
  -- Contains one or more presentation data values from the same
  -- presentation context.
  -- See 8.4.2.
}

User-session-requirements ::= BIT STRING {
  half-duplex(0), duplex(1), expedited-data(2), minor-synchronize(3),
  major-synchronize(4), resynchronize(5), activity-management(6),
  negotiated-release(7), capability-data(8), exceptions(9), typed-data(10),
  symmetric-synchronize(11), data-separation(12)}

```

END

-- Generated by Asnp, the ASN.1 pretty-printer of France Telecom R&D